Use-Case Flow Specification- *Micro-Lending & Peer Credit Risk Assessor*

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Use Case: UC-07 Make Repayment

Primary Actor: Borrower

Secondary Actor: Payment Gateway (external system, via included use case UC-08 : Process Payment)

Related Use Cases: «include» UC-08 Process Payment; «extend» by UC-09 Trigger Default Warning

Preconditions

- The borrower is authenticated and has an active, funded loan with a repayment schedule generated (UC-06).
- At least one EMI installment on the schedule is currently due or overdue.
- The borrower has a valid, linked payment method (bank account, UPI, or card).

Postconditions (Success)

- The EMI amount is deducted from the borrower's selected payment method and settled to the peer lender's account.
- The corresponding installment on the repayment schedule is marked "Paid" and the outstanding balance is recalculated.
- A payment confirmation receipt is generated and made available to the borrower; the peer lender's portfolio view is updated.

Main Success Scenario

1. Borrower selects "Make Repayment" from the loan dashboard.

2. System displays the current outstanding EMI amount, due date, and remaining balance.

3. Borrower selects a payment method (linked bank account, UPI, or card).

4. System validates the repayment amount against the repayment schedule.

5. System invokes UC-08 Process Payment «include», initiating the transaction with the Payment Gateway inside an ACID-compliant database transaction (per NFR-001).

6. Payment Gateway authorizes and confirms the transaction.

7. System updates the repayment schedule, marking the installment "Paid" and recalculating the outstanding balance.

8. System generates and sends a payment confirmation receipt to the borrower and refreshes the peer lender's portfolio view.

9. Use case ends successfully.

Alternate Flow 6a: Payment Declined or Missed

- 6a1. The Payment Gateway declines the transaction, or the EMI due date passes with no payment attempt.
- 6a2. System logs the failed/missed attempt and prompts the borrower to retry with the same or an alternate payment method within a 3-day grace period.
- 6a3. If no successful payment is received within the grace period, the system triggers UC-09 Trigger Default Warning «extend», notifying both the borrower and the peer lender.
- 6a4. Use case ends in the "Default Warning Issued" state; the installment remains "Overdue" until resolved.